GENERAL SPECIFICATION G1.3

Workmanship Requirements

CONTENTS

G1.3 WORKMANSHIP REQUIREMENTS

G1.3.1 Scope

(a) This Part of the Specification sets out the required minimum general standards for materials, workmanship, design and testing for plant to be adopted by the Contractor, and reference to any specific material or equipment does not necessarily imply that such material or equipment is included in the Works.

(b) All relevant component parts of the Works shall, unless otherwise specified, comply with the provisions herein and be subject to the approval of the S.O..

G1.3.2 Plant Design - General

(a) The Plant shall be new, of sound workmanship and robust design, and of a grade and quality entirely suitable for the climatic and working conditions at the Site. As the plant site is in an environment at close proximity to the sea, the corresponding environmental factors aggravating corrosion shall be taken into design consideration through appropriate material selection, protective coatings or painting, cladding, corrosion allowance, cathodic protection, etc. In addition, tests for soil aggressiveness shall be conducted as far as necessary to determine the suitability of materials of construction for buried entities such as pipes, valves, etc. All proposed materials of construction and measures of protection against corrosion for implementation shall be submitted to the S.O. for approval.

(b) For the purposes of design, the Plant shall be suitable for operating throughout the ranges of conditions described in the Specification. Due attention shall be given to expansion due to temperature changes, the stability of paint finish for high temperatures and high relative humidity, the rating of engines, electrical machinery, thermal overload devices, cooling systems, and the choice of lubricants for the possible prolonged high operating temperatures.

(c) The Plant shall be designed to provide protection against and avoid damage by the entry of vermin and dust, and to minimise fire risk and consequent fire damage. Plant shall be protected against damage due to dampness and condensation by sealing or fitting anti-condensation heaters of suitable rating.

(d) Suitably designed lightning surge protection devices shall be incorporated into any electronic equipment, if the manufacturer of the equipment deems it to be necessary.

(e) All manually controlled Plant located outdoors shall be provided with facilities for making it tamperproof. This is in addition to any requirements of the Specification for securing Plant under operational conditions.

(f) All component parts of the Plant shall be manufactured to strict limits of accuracy and shall be interchangeable with the component parts of similar Plant.

(g) The Plant shall be designed for continuous operation for prolonged periods with a minimum of maintenance and shall have a high resistance to change in these properties due to passage of time, exposure to light, vibration and noise or any other cause, which may affect the performance or life of the Works. The Contractor may be called upon to demonstrate this for any component part either by service records of similar equipment, or by the records of extensive type tests.

(h) Materials shall be selected taking into consideration their location and duty as well as specifications. In the case of Plant conveying water, particular attention shall be given to the risk of electrolytic reaction between differing materials of construction and to the effects of corrosion and, where there are impurities in the water may cause erosion.

(i) Where wear is likely to occur during normal operation, the Plant shall be designed to enable a potentially affected area of a component part to be replaced without replacing the whole component. No part subject to wear shall have a life from new to replacement or repair of less than one year of continual operation. Where major dismantling to replace a part cannot be avoided, the life of such parts shall not be less than 5 years.

(j) The following parts shall have lives of not less than the number of years stated below:

Equipment parts	**Years**
Pumps	
Impeller	10
Neck Rings and bushes	10
Gland sleeves – if replaceable without dismantling the pump	3
Gland sleeves – if replaceable with the need to dismantle the pump	10

Other Parts	**Years**
Sleeve bearings	10
Ball and roller bearings	5
Control valve discs and seats	5
Other valve discs and seats	10
Spindles, axles and shafts	20
Open gearing, chains, sprockets, etc.	10
Gearing in boxes	20
Wearing parts of screens, strainers, etc.	10
Reciprocating parts of motors, compressors, etc.	10
Gland packing except initial packing	1

(k) Electrical contractors and switches shall have a minimum life under normal full load working conditions of two million operations without replacement of any part and two hundred thousand operations without maintenance of any kind.

(l) Electrical relays and light current switching devices shall not require maintenance of any kind before completing two million operations under normal full working load.

(m) Where guarantees of the life of membranes or other plant are included in the contract, such guarantees shall prevail.

G1.3.3 Substances and Products

Substances and products used in the Works, which may be applied to or introduced into water which is to be supplied for drinking, washing or cooking shall not contain any matter which could impart taste, odour, colour or toxicity to the water or otherwise be objectionable on health grounds.

G1.3.4 Materials

(a) All materials incorporated in the Works shall be the most suitable for the duty required and shall be new and of first class commercial quality, free from imperfections and selected for long life and minimum maintenance. They shall be specifically chosen as suitable for use in the hot and humid tropical climate of Singapore and for deployment at the project Site, which is coastal.

(b) All submerged moving parts of the Plant, or the pins and spindles, etc., of the submerged moving parts or the faces, etc., in contact with them shall be of non-corrodible metals. All parts in direct contact with chemicals shall be completely resistant to corrosion or abrasion by those chemicals and shall also maintain their properties without ageing due to the passage of time, exposure to light, changes in ambient or operating temperature and pressure or other environmental conditions.

(c) Particular attention shall be paid to the prevention of corrosion due to the close proximity of dissimilar metals.

(d) When it is necessary to use dissimilar metals in contact, these shall be so selected that the potential difference between them in the Electro chemical series is not greater than 0.5 milliVolt (mV). If this is not possible, the contact surfaces of one or both of the metals shall be electroplated or otherwise finished in such a manner that the potential difference is reduced to within required limits or, alternatively, the two metals shall be insulated from each other by an approved method. In addition, attention shall also be paid to the prevention of seizure by fretting where two corrosion resistant metals are in contact, by the selection of materials of suitable relative hardness and surface finish and the application of lubricants. Where bronze is specified or used, it shall be zinc free.

G1.3.5 Workmanship

(a) Workmanship and general finish shall be of first class commercial quality and in accordance with best shop practice and shall be performed by people skilled in their respective trades.

(b) All similar items of Plant and their component parts shall be completely interchangeable. Spare parts shall be manufactured from the same materials as the originals and shall fit all similar items of Plant. Machining fits on renewable parts shall be accurate and to specified tolerances so that replacement of parts according to manufacturer's drawings may be readily installed.

(c) All equipment shall operate without excessive vibration and with a minimum of noise. All revolving parts shall be truly balanced both statically and dynamically so that when running at normal operating speeds and at any load up to the maximum, there shall be minimal vibration.

(d) Where specified, limits of vibration of rotating equipment shall be within the limits for the class of the machine as specified in the relevant BS EN (British Standard European Norm) Standards.

(e) All parts, which can be worn or damaged by dust, shall be totally enclosed in dust proof housings.

(f) All equipment shall be designed to prevent malfunction due to rust or corrosion, to minimise risk of fire and to be electrically and mechanically safe to operators.

G1.3.6 Material Samples

(a) Where the Contract requires the submission of samples, they shall be submitted by and at the expense of the Contractor not less than thirty calendar days prior to that time that the materials represented by such samples are needed for incorporation into any work. Samples shall be subject to approval by the S.O., and material represented by such samples shall not be manufactured, delivered to the Site or incorporated into any work without such approval.

(b) The S.O.'s approval of a sample of material for incorporation into the works shall not be taken as approval for use of such material at specific locations on the works. Only when approved drawings are returned to the contractor for particular areas of the works shall the use of approved materials be deemed acceptable.

(c) Separate payment will not be made for complying with the requirements of this clause and all costs shall be deemed to be included in the amounts contained in the Schedule of Prices. All material samples submitted by the Contractor will be retained by the S.O. but shall ultimately be disposed off by the Contractor, when requested by the S.O..

(d) A sample of any material, once approved for incorporation into the Works and retained by the S.O. will be used as a yardstick by which the S.O. will judge the standard of manufacture and installation of the material represented by the sample incorporated in the works. The materials and workmanship exhibited in the approved sample shall represent the minimum acceptable quality for incorporation into the works. Consequently, upon delivery to site or installation of quantities of the approved materials (as represented by the samples), which the S.O. considers to be of a quality inferior to the approved sample, the concerned materials shall be removed from site and replaced with materials of quality at least as good as the approved sample.

G1.3.7 Allowance for Wastage

(a) The Contractor shall provide as specified and to the satisfaction of the S.O. reasonable excess quantities to cover wastage of those materials which will be normally subject to waste during erection, setting to work, testing and commissioning.

G1.3.8 Tropicalisation of Electrical Components

(a) The following general requirements shall be observed:

(i) Unpainted steel parts - Passivated cadmium plated galvanised or zinc plated as appropriate;

(ii) Screws, nuts, bolts - The use of steel shall be avoided where possible or when used shall be cadmium, zinc or chromium plated. All non-ferrous screws shall be electro tinned, or shall have nickel or chromium plate finish;

(iii) Springs - Springs where possible shall be of phosphor bronze or nickel silver;

(iv) Wood - All wood shall be seasoned primary hardwood;

(v) Insulating materials - None impregnated paper, fabric, wood or presspahn shall not be used for insulating purposes. Where synthetic resin bonded insulating boards are used, all cut edges shall be sealed with an approved varnish;

(vi) Instruments, relays - All instruments, relays, etc. shall be provided with tropical finish;

(vii) Operating coils - Where practicable all fine wire operating coils and wire wound resistors shall be vacuum impregnated with an approved insulating varnish; and

(viii) Transformer windings - an approved insulating varnish, epoxy resin encapsulating varnish, epoxy resin encapsulated or protected against the ingress of moisture in an approved manner.

G1.3.9 Erection Marks

(a) Before dismantling for packing, all shop assemblies shall be clearly scribed with reference lines and match marked to facilitate correct reassembly.

(b) All members comprising multi-part assemblies shall be marked with distinguishing numbers and/or letters corresponding to those on the drawings or material list provided. Erection marks impressed before painting or galvanising shall be clearly readable thereafter.

G1.3.10 Working Stresses

(a) Except as otherwise specified under the provisions of Standards or of the Specification the limitation of working stresses shown in the following table shall apply under maximum operating conditions.

	Tension	**Compression**	**Shear**
Grey Cast Iron	One-tenth of Ultimate strength	70 MPa	21 MPa
Carbon Cast Steel	83 MPa	83 MPa	50 MPa
Alloy Cast Steel	One-fifth of Ultimate Strength or one-third of Yield Strength (whichever is the lesser)	One fifth of Ultimate Strength or one third of Yield Strength (whichever is the lesser)	60 %% of allowable tensile stress

	Tension	Compression	Shear
Plate Steel	One quarter of Ultimate Strength but not to exceed 103 MPa	One quarter of Ultimate Strength but not to exceed 103 MPa	60% of allowable tensile stress

(b) For other materials used, the working stresses in tension or compression due to the most severe conditions shall be one third (1/3) of the yield strength or one fifth (1/5) of the ultimate strength of the material, whichever is the lesser. Upon request by the S.O., complete information regarding maximum unit stresses used in the design shall be provided by the Contractor.

G1.3.11 Steel to be Welded

(a) All cast steel parts to be welded shall be manufactured in steel produced by the open hearth or electric processes, acid or basic processes, with an ultimate strength in the range 430/510 MPa and a carbon content not higher than 0.25%. It shall show on analysis a phosphorous content of not more than 0.05%. The steel used shall not require preheating for efficient welding and shall have high resistance to "Notch Brittleness".

G1.3.12 Welding

(a) In all cases where welds are liable to be highly stressed, the Contractor shall provide to the S.O. before fabrication commences, detailed drawings of all weld preparations and procedures proposed. No such welding shall be carried out before the S.O. has approved procedures. No alteration shall be made to any previously approved procedures without prior approval of the S.O.. Welders shall be qualified in accordance with the requirements of the appropriate part of ISO 15614-8: 2016 B. Welders shall be required to be tested at Site by the S.O. to confirm they possess the required welding skills.

G1.3.13 Casting

(a) The structure of the castings shall be homogeneous and free from non-metallic inclusions and other defects. All surfaces of castings, which are not machined, shall be fettled to remove all foundry irregularities.

(b) Minor defects not exceeding 10 mm in depth or 10% of total metal thickness whichever is less or which will not ultimately affect the strength and serviceability of the casting may be repaired by welding. The Contractor shall notify the S.O. of larger defects and no repair welding of such defects shall be carried out without prior approval of the S.O..

(c) If the removal of metal for repair would reduce the stress-resisting cross-section of the casting by more than 25%, or to such an extent that the computed stress in the remaining metal exceeds the allowable stress by more than 25%, then that casting shall not be used in the Works.

(d) Castings repaired by welding for major defects shall be stress-relieved after such welding, or as otherwise instructed in writing by the S.O..

(e) Non-destructive tests shall be carried out for any casting containing defects whose effect cannot otherwise be established, or to determine that repair welds have been properly made. The method of non-destructive tests shall be consented by the S.O..

(f) Unless otherwise specified castings shall be produced to the following standards:

Flake graphite cast iron	BS EN 1561
Carbon steel, Stainless Steel	BS EN 10293
Copper & copper alloy	BS EN 1982

G1.3.14 Non-Metallic Materials

(a) Fabrics, cork, paper and similar materials, which are not subsequently to be protected by impregnation, shall be treated with a fungicide. Sleeving and fabrics treated with linseed oil varnish will not be permitted.

(b) The use of organic materials shall be avoided as far as possible but where these have to be used, they shall be treated to make them fire resistant and non-flame propagating.

(c) The use of wood shall be avoided as far as possible. If used, woodwork shall be thoroughly seasoned teak or similar hardwood which is resistant to fungal decay and other blemishes. All woodwork shall be treated to protect it against damage by fire, moisture, fungus, vermin, insect, bacteria or chemical attack, unless it is naturally resistant to all these. All joints in woodwork shall be dovetailed or tongued and pinned. Metal fittings on wood shall be of non-ferrous material. Adhesives shall be impervious to moisture and fungus growth. Synthetic resin cement only shall be used for joining wood. Casein cement shall not be used.

G1.3.15 Engineering Hardware

(a) Nuts and bolts for pressure fittings shall be of high quality stainless steel, machined on the shank and under the head and nut. Bolts shall be of such a length that only one to three threads shall show through the nut when in the fully tightened condition.

(b) Bolts shall be selected as recommended by the manufacturer for the application. Under no condition, shall incorrectly selected bolts (in terms of length, diameter or materials) shall be used.

(c) Fitted bolts shall be a light driving fit in the reamed holes they occupy, shall have the screwed portion of such a diameter that it will not be damaged in driving and shall be marked in a conspicuous position to ensure correct assembly at site.

(d) Washers, locking devices and anti-vibration fittings shall be provided where necessary to ensure that no bending stress is caused in the bolt.

(e) When there is a risk of corrosion, bolts and studs shall be designed so that the maximum stress in the bolt does not exceed half the yield stress of the material under all conditions.

(f) All bolts, nuts and screws which are subject to frequent adjustment or removal in the course of maintenance and repair shall be made of nickel-bearing stainless steel or brass.

(g) All bolts, nuts and screws and washers used on submerged equipment shall be stainless steel.

(h) The Contractor shall provide all holding down, alignment and levelling bolts complete with anchorages, nuts, washers and packings required to attach the Plant to its foundation, and all bedplates, frames and other structural parts necessary to spread the loads transmitted by the Plant to concrete foundations without exceeding the design stresses.

(i) Isometric black hexagonal bolts, nuts and screws shall comply with BS 4190:2014 strength grade 4.6.

(j) Metal washers for general purposes shall comply with BS 4320.

(k) Isometric precision hexagonal bolts, nuts and screws shall comply with BS 3692 strength grade 8.8.

(l) High strength friction grip bolts, nuts and washers shall comply with BS EN 14399-1-10.

(m) Black cup and countersunk head bolts and screws with nuts shall comply with BS 4933.

(n) Stainless steel nuts, screws, washers and bolts shall be manufactured from Grade 316S31 steel complying with BS EN 10095.

(o) Bolting for pipes and fittings shall comply with the relevant provisions of BS EN 1092 except that spheroidal graphite iron bolts for use with ductile iron pipes and fittings shall be manufactured from metal complying with the provisions of BS EN1563 for grade 500/7.

G1.3.16 Threads

(a) All threads shall be of metric sizes with the standard coarse form of medium fit to BS 3643 except for special applications for which the metric fine thread may be utilised, or other thread forms subject to the approval of the S.O..

G1.3.17 Instrument Screws, Nuts, Springs and Pivots

(a) The use of iron and steel screws shall be avoided in instruments and electrical relays wherever possible.

(b) Steel screws, when used, shall be zinc, cadmium or chromium plated or, where plating is not possible due to tolerance limitations, of stainless steel. All wood screws shall be of dull nickel-plated brass or of other approved finish. Instrument screws (except those forming part of a magnetic circuit) shall be of brass or bronze. Springs shall be of non-rusting materials e.g. phosphor bronze or nickel silver, as far as possible. Pivots and other parts for which non-ferrous material is unsuitable shall be of an approved stainless steel.

(c) Iron and steel shall in general be painted or hot dip galvanised as appropriate in accordance with the Specification. Indoor parts may alternatively have chromium or copper nickel plated or other approved protective finish.

(d) Small iron and steel parts (other than stainless steel) of all instruments and electrical equipment, the poles of electromagnets and the metal parts of relays and mechanisms shall be chromium or copper nickel plated or have some other approved finish to prevent rust. Cores and the like which are built up of laminations or cannot for any other reason be anti-rust treated, shall have all exposed parts thoroughly cleaned and heavily enamelled, lacquered or compounded.

G1.3.18 Rivets

(a) For general use, pan head rivets shall be used. Rivets on bearing surfaces shall be flat counter sunk, driven flush.

(b) Loose rivets, or rivets with heads badly formed, cracked or eccentric to the shank, which do not bear truly on the plate or bar shall be replaced.

(c) All surfaces to be riveted shall be in intimate contact throughout.

G1.3.19 Safeguarding Plant

(a) The Contractor shall ensure that all design and installation of all equipment for which he is responsible does not present any danger of injury or death to personnel. Nothing in this Specification shall remove the Contractor's obligation of drawing to the attention of the S.O., to any feature of the Works which is not consistent with safety, or to prevent him from making proposals for incorporating equipment or designs, which would increase the safety of the Plant.

(b) The installation layout and plant design shall not allow any item of plant to be so positioned that danger to operating personnel could arise during normal operation and maintenance. Particular attention shall be paid to the safe positioning of pipes, valve handwheels, air vents and rotating machinery.

G1.3.20 Plant and Equipment Identification and Tagging

(a) General

(i) All plant and equipment shall be clearly identified with both a physical metal tag and a Radio-frequency identification (RFID) tag. Where required equipment shall be provided to print RFID tags for installed plant, equipment and inventory spares and materials.

(ii) Physical Tags

1) The Contractor shall ensure that each main and auxiliary item of Plant and equipment shall have permanently attached to it in a conspicuous position, a nameplate and rating plate (tag). Upon these shall be engraved the manufacturer's name, direction of rotation, type and serial number of plant, details of the loading and duty at which the item of Plant has been designed to operate, and such diagrams as are deemed necessary. All indicating and operating devices shall have securely attached to them or marked upon them designations as to their function and proper manner of use. Provision shall be made to incorporate descriptive numbering codes (Tag Nos.) as indicated on the layout drawings.

2) A nameplate shall be provided and mounted on or adjacent to each piece of chemical feed equipment to identify its function.

3) Feeder designations on the nameplates shall correspond to those indicated on the drawings.

4) All valves shall have an identification plate bearing the valve number and a short description of valve function.

5) On major items of plant and valves, details of proposed plates, labels and inscriptions shall be provided by the Contractor for approval by the S.O..

6) Such nameplates, rating plates and labels shall be of a non-flame propagating material, either non-hygroscopic or transparent plastic, with engraved lettering of a contrasting colour. Fixing shall be by means of screws. No drive rivets or adhesives shall be used.

7) Nameplates shall be made from 2 mm thick SS316 material.

8) Nameplates shall bear 10 mm high engraved block type black enamel filled equipment identification numbers and letters.

(iii) The nameplate shall state, as a minimum, the following:

1) Manufacturer and date of manufacturer;

2) Model Number;

3) Serial Number;

4) Power rating for the drive; and

5) Major Component Weights.

(b) Barcode and RFID Label Printer

(i) Barcode and RFID label printers shall meet the following minimum requirements:

1) Supports barcode printing and RFID encoding on UHF Passive Label provided in this Contract;

2) Supports printing over the Board’s network via 802.1X protocols; and

3) Printer driver/software available/supported in 64 bit and Microsoft Windows 10 client’s environments.

G1.3.21 Floor Collars and Wall Boxes

(a) At all locations where pipes pass through concrete floors and walls or brick walls or other walls and the pipes are not to be built in, suitable floor collars or wall boxes shall be designed by the Contractor. Where pipes pass through firewalls or slabs, collars shall be approved fire collars.

(b) The floor collars shall have raised kerbs of suitable height, which shall not be less than 75 mm. The wall boxes shall be flush fitting and of neat design and approved finish.

(c) The Contractor shall provide all fittings for the passage of pipes through external walls together with the provider of all components for the boxing in of the holes against inclement weather.

(d) After the collars and boxes or other fittings have been fixed in position, the floor, walls and roof structures shall be made good by the Contractor.

(e) Where service pipes run adjacent to each other, they shall, wherever possible, pass through a common box. Where pipes of varying bore pass through a common box, a neat plate cover shall be provided and fixed.

(f) In the case of flanged pipework, boxes shall be large enough to permit the passage of the flange. Under no circumstances shall flanged connections be built into walls or floors. Flanges adjacent to floors and walls shall be sufficiently distant to allow access for removal of gaskets.

(g) Where pipes or conduits are to be built into concrete which may be subject to differential water pressure, puddle flanges shall be fitted of such size and number to seal the leakage path and shall be capable of carrying all operation longitudinal forces falling on the pipe or conduit without reliance on adhesion of the pipe surface and without exceeding the bearing pressures normally allowed on concrete.

(h) Where pipes pass through chequer plate or open mesh flooring, suitable floor collars of an approved type shall be provided and fixed by the Contractor.

G1.3.22 Hoisting

(a) Where ropes are used for temporary or permanent purposes for hoisting or winching, they shall be of multi core steel wire ropes, heavily galvanised or otherwise suitably protected from corrosion.

G1.3.23 Aluminium Structures

(a) Aluminium structures shall be constructed in alloy NE8M to BS EN 573 Part 3:2013. Cold working shall be avoided.

(b) Aluminium structural members shall comply with BS EN 755:2013, Parts 1 – 9.

(c) The minimum thickness of any member shall be 8 mm.

(d) Where the aluminium members are in contact with structures other than aluminium structures, the contact faces shall be protected by insulating pads of neoprene bitumen alloy.

(e) All exposed component of aluminium structure shall be anodised to an approved colour.

G1.3.24 Lubrication

(a) General

(i) Provision shall be made for suitable lubrication to ensure smooth operation, heat removal and freedom from undue wear. Equipment selected shall require minimum lubrication maintenance and down time for lubricant change.

(ii) The Contractor shall provide the first fill of oil and grease from approved lubricant suppliers.

(iii) All grease nipples, oil cups and dip sticks shall be readily accessible, being piped to a point as near as practicable to the lubrication point.

(b) Oil Lubrication

(i) Gear boxes and oil baths shall be provided with adequately sized filling and draining plugs and suitable means of oil level indication.

(ii) Roller chain drives shall have oil bath reservoir lubrication.

(iii) Drain points shall be located or piped to a position such that an adequately sized container can be placed beneath them. Where a large quantity of oil is involved or drainage to a container difficult, a drain valve and plug shall be provided at the point of discharge.

(iv) Bearings equipped with forced fed oil lubrication shall be automatically charged prior to machinery starting up and pressure monitored during operation with automatic shutdown of machinery and alarm on low oil pressure.

(v) All points where oil leakage may occur shall be suitably trapped to prevent oil contamination of water. Oil filling and drain points shall be arranged so as to avoid the risk of contamination of water by accidental spillage.

(vi) Access provision, without the use of portable ladders, to lubrication systems shall be such as to permit maintenance, draining and re-filling, without contamination of the charged lubricant.

(vii) The design of breathers shall take into account the humidity and atmospheric contamination at the vent point and measures shall be incorporated to prevent contamination of the lubricant.

(c) Grease Lubrication

(i) Grease application shall be by steel lubrication nipples.

(ii) Anti-friction bearings requiring infrequent charging shall be fitted with hydraulic type nipples.

(iii) Plain bearings requiring frequent charging shall be fitted with button head pattern nipples.

(iv) A separate nipple shall be provided to serve each lubrication point. Where a number of nipples provide remote lubricating points, they shall be grouped together on a conveniently placed battery plate, with spacing in accordance with the recommendations of BS 1486, Part 1, Tables 9 and 10.

G1.3.25 Gaskets and Joint Rings

(a) Joint rings suitable for hot or cold water or specified hydrocarbon fluids or for drainage or sewerage applications shall be manufactured to conform with BS EN 681-1, BS EN 681-2 and BS EN 682 and shall be of chloroprene rubber or other approved synthetic material suitable for temperatures up to 80 deg. C., or greater to suit the application.

(b) Joints shall be made in accordance with manufacturer's instructions or as specified herein.

(c) Until immediately required for incorporation in a joint, each rubber ring or gasket shall be stored in the dark free from the deleterious effects of heat or cold and kept flat so as to prevent any part of the rubber being in tension.

(d) Only lubricants recommended by the manufacturer shall be used in connection with rubber rings and these lubricants shall not contain any constituent soluble in water of the quality stated in the Specification, shall be suitable for the climatic conditions at the Site and shall contain an approved bactericide.

(e) Graphite grease or similar shall be applied to the threads of bolts before joints are made.

G1.3.26 Filling Materials

(a) The Contractor shall provide all packing, filling compounds, jointing materials, insulating and cooling fluids and the like needed for the completion of the Works.

G1.3.27 Electroplating, Galvanising and Sherardizing

(a) Hot dip galvanising shall be carried out in accordance with BS EN ISO 1461 with a deposition rate of at least 460 g/square metre. After galvanising all parts shall be passivated to minimise discoloration.

(b) Electroplating or galvanising will be acceptable as an alternative to painting for small ferrous components.

(c) All fixing bolts, washers, nuts and other fixings required for erection shall be spun galvanised or sherardized in accordance with ASTM B695-04 and ASTM B633-15 unless otherwise specified.

G1.3.28 Control of Vibration

(a) All items of rotating machinery shall be dynamically balanced so that the level of vibration is within the limits set by BS ISO 10816-1, for a class IV machine to grade B.

(b) This limit on vibration shall extend to the communicating pipework and mounting arrangements and include adjacent machinery either in operation or not.

G1.3.29 Access Steelwork, Stairs and Walkways

(a) Any areas of chequer plating or similar covering that are necessary to cover gaps between items of Plant and the surrounding structure, and any access ladders, platforms and handrails that must be attached to items of Plant to facilitate operation, inspection or maintenance, shall be provided by the Contractor to the satisfaction of the S.O..

(b) The Contractor shall provide adequate access to all handwheels, sight glasses, gauges, lubrication points and any other items to which access is necessary for routine maintenance, and across pipework in pipe galleries and basements.

(c) Access platform shall be provided for all plant and equipment which is 1.5m above the floor level for safe operation and maintenance.

(d) Chequer plating shall be of aluminium 'Durbar' or other non-slip pattern, not less than 4.5 mm thick (exclusive of pattern). Diamond type pattern chequer plate shall not be used. Open type or solid type chequer plate flooring shall be used as appropriate for the location, taking into account ease of cleaning, precautions against slipping and areas below walkways.

(e) During tender design, the S.O. has indicated the location and general arrangement of access stairs, platforms and walkways in the preliminary BIM model. The Contractor shall develop designs and provided additional steelwork, stairs, platforms and walkways needed for safe access and allow easy maintenance of equipment. The cost of such will be deemed to be included in the Contract Price. No additional payment will be made for the provision of such facilities, not included by the S.O. at the time of Tender.

G1.3.30 Handrailing

(a) All handrailing shall be of a single design compliant with the relevant standard details provided in Volume 5B. Where vendor equipment standard designs of handrailing differs from those specified, approval for deviation shall be sought form the S.O., who will consider this on a case by case basis.

G1.3.31 Machinery, Lifting and Dismantling

(a) Machinery bedplate design, packing and fixing shall be such as to minimise distortion and vibration. Aligned machinery shall be mounted on either bed or sole plates, permitting removal and reinstatement without a requirement to re-grout. Bedplates shall incorporate fine adjustment of the vertical and horizontal alignment between driver and driven members.

(b) All machinery shall be fitted with lifting facilities. Large structures shall be provided with jacking points.

(c) Tapped holes or other provision must be made in all main castings, for the insertion of jacking screws or the fixing of drawing gear to facilitate dismantling on items of machinery subject to frequent dismantling. Bolts or studs shall be employed in preference to set screws.

G1.3.32 Seals

(a) General

(i) The Contractor shall select seals, compatible with the Plant and best suited for the worst conditions likely to be met when the Plant is in operation.

(ii) All seal materials shall be compatible with and/or resistant to the fluid or gas being handled. For potable water, seal materials shall be specifically approved.

(b) Soft Packed Glands

(i) Shafts shall be provided with renewable gland sleeves. Glands subject to abrasive liquors or negative pressures shall embody suitably positioned lantern rings and a clean water continuous flushing system, operative whenever the Plant is in motion.

(ii) Gland adjustment nuts shall be readily accessible for routine maintenance.

(iii) Gland drain pipework, shall be installed, incorporating rodding facilities and adequate inclines, of 25mm minimum diameter on water reclamation plant and 12.5mm on water supply plant, discharging to the nearest sump or drainage channel.

(c) Mechanical Seals

(i) Mechanical seals which are subject to abrasive liquor or gas, negative pressures or corrosive elements, shall be provided with a clean water continuous gland flushing system, operative when the item of plant is in motion or the corrosive element present. A back-to-back sealing arrangement with a flushing/cooling system shall be accepted as satisfying the requirements of this clause.

G1.3.33 Bearings

(a) Below Water Bearings

(i) The Contractor shall select the most appropriate types of bearings for the Plant being supplied.

(ii) Equipment with vertical shafts shall have thrust and guide bearings. All bearings shall be designed to exclude the ingress of dust and water.

(iii) Sealed for life units are preferred subject to a minimum design life of 50,000 hours operation at maximum loading.

(iv) Plant which may be subject to vibration whilst stationary shall be provided with bearings designed to withstand damage from such a cause.

(v) Below water bearings shall be of the journal type, of ferrobestos, gunmetal or equivalent and journals of stainless steel.

(b) Above Water Bearings

(i) Wherever practicable, bearings shall be of the sealed for life type.

(ii) Single journal plain bearings shall be phosphor bronze or synthetic lubrication impregnated bushes with carbon or stainless steel journals respectively. Synthetic bearings shall only be used where bearing condition can be inspected readily.

(iii) Plain type bearings shall be self-lubricating by either grease, forced oil or impregnation.

(iv) Ball and roller type bearings shall be adequately lubricated by oil or grease and sealed to prevent leakage of lubricant along the shaft. Attention shall be given to ensure the dismantling of bearings is simple and free from risk of damage.

(v) Bearings fitted to gear boxes shall a minimum design life of 100,000 hours at maximum loading.

G1.3.34 Gearboxes

(a) Gearboxes shall have a minimum life of 100,000 hours, be selected in accordance with the American Gearbox Manufacturers' Association recommendation for horsepower (or kW) calculation and service factor application and employ a standard reduction ratio.

(b) Gearboxes, which have to be angle-mounted, shall have a rating, choice of bearings, seals and lubrication system which are suitable for such mounting.

(c) Dependence on splash lubrication alone is not acceptable but it may be used in conjunction with a forced feed method to reach all bearings and gears.

(d) Calibration of the oil dipstick and its position together with that of the sump drain plug will require special consideration.

G1.3.35 Safety Signs

(a) All signs providing health and safety information or instructions shall comply with BS 5499.

(b) The signs shall be of durable quality and shall comprise a substrate of 22 gauge aluminium, pre drilled for fixing and with radiused corners free of burrs or sharp edges. Symbols and lettering shall be screen printed.

G1.3.36 Machine Guards

(a) All moving parts shall be protected by machine guards.

(b) For the purposes of this Specification, a machine guard is defined as a cover which is fitted over or adjacent to a moving part of the equipment (for example, a shaft coupling, a gear or belt drive, a counterweight or moving linkages) and shall comprise a casing (with or without a frame) and the necessary mounting fasteners.

(c) Machine guards shall conform to British Standard Institute advisory PD 5304 and all other statutory requirements.

(d) Guards shall:

(i) Be made of sheet metal or chain mesh;

(ii) Be supported so as to minimise its vibration;

(iii) Be capable of ready removal and replacement;

(iv) Not restrict access to bearings or other items requiring inspection or maintenance;

(v) For guards located in areas of limited access, be of sectional construction and so arranged that access for maintenance can be obtained by manual removal of a number of sections;

(vi) Be provided with removable cover plates, where relevant, to provide access to overload protective devices specified so as to allow replacement or resetting of these devices without the use of tools;

(vii) Be fitted with a hinged inspection door, secured with wing nuts or dog cleats in the case where the guard covers a belt or chain to enable in service inspection of the drive without removing the guard; and

(viii) Be capable of withstanding a pressure of 75 kPa from any direction without permanent distortion.

(e) Guards for personnel protection shall be provided for:

(i) All exposed couplings, gears and belt or chain drives;

(ii) Moving counterweights and levers;

(iii) As set out in PD 5304 and statutory regulations; and

(iv) Other moving parts as required by the S.O..

G1.3.37 Locks

(a) Where locks are specified or provided, they shall be of the cylinder type and three keys shall be provided for each lock. Where locks are provided for a particular group of items (e.g. instrument cabinet) keys shall be interchangeable.

(b) Master keys shall be provided for groups of essential locks.

(c) Locks and padlocks shall be made of brass.

(d) Lockable keyboards of approved design shall be provided for the storage of groups of keys and padlocks for particular items, or groups of items of plant when not in use.

(e) All locks and keys shall be clearly engraved or be provided with engraved labels stating their purpose and the keyboards shall be provided with labels, identically engraved to enable each key and padlock to be identified and located.

END OF SECTION